Kinetic energy and net work

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Learning target: Connect net work to a change in kinetic energy.

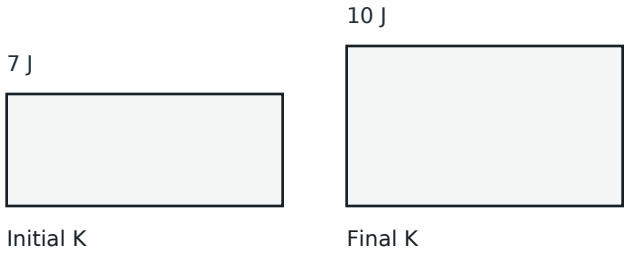
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 1.*
Factor 9, because $K=\frac{1}{2}mv^2$.

02. Worked example: explain the model
Bars 4 and 9 J; +5 J net work. Speed is not increased by 5 m/s.

03. Read and interpret the model
Net work = 10 - 7 = +3 J. Its speed increases.

Model / diagram check



04. Apply the physics
9 J; final 16 J.

05. Explain your reasoning
Negative friction work reduces net work and final kinetic energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 6.*
Yes: sticking defines perfectly inelastic. Total momentum is conserved; kinetic energy need not be.

07. Exit check
-12 J; speed falls.

08. STRETCH: test the model
Changes 4-1=3 J and 9-4=5 J; equal speed increases require different energy changes.

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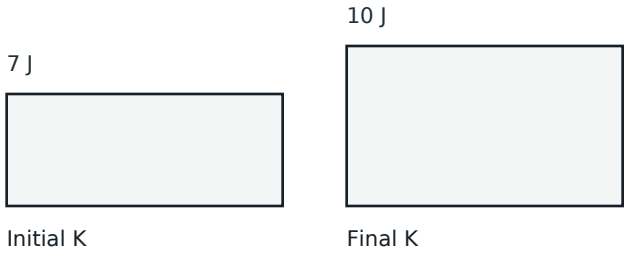
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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 1.*
K becomes 1/3 of the original; fixed p means K∝1/m.

02. Worked example: explain the model
Bars 4 and 9 J; +5 J net work. Speed is not increased by 5 m/s.

03. Read and interpret the model
Net work = 10 - 7 = +3 J. Its speed increases.

Model / diagram check



04. Apply the physics
Initial 8 J; final 32 J; v=√(64/4)=4 m/s.

05. Explain your reasoning
Negative friction work reduces net work and final kinetic energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 6.*
A stops; B moves at +6 m/s. Equal masses exchange velocities in an ideal 1 D elastic collision.

07. Exit check
Final 4 J-initial 36 J=-32 J.

08. STRETCH: test the model
Initial 24 J, final 6 J, v=2 m/s; stopping work -24 J.